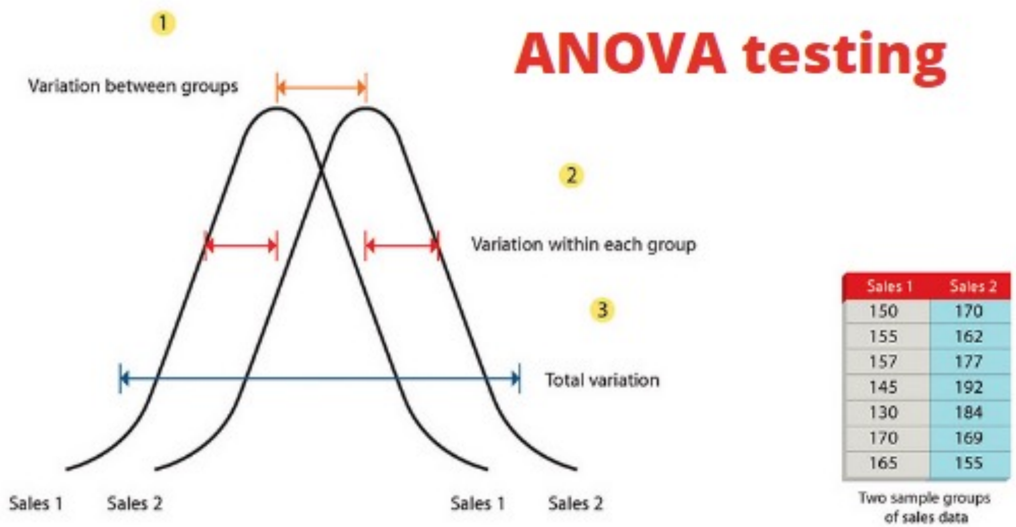
# Week 7 – Seminar: ANOVA



**Recall from lectures…**

**ANOVA**: stands for **Analysis of Variance**. It is essentially a measure of the goodness of fit of a model.

- Assume we would like to test whether all the colleges have the same mean simultaneously and have an overall significance level of 5%. This is the ANOVA test.
- The ANOVA test considers the variation between the groups involved compared to the variation within the groups.
- <u>ANOVA follows the same thought process as before:</u>

| Height | College |
|---|---|
| 1.56 | A |
| 1.78 | V |
| 1.54 | C |
| 1.70 | D |
| 1.70 | J |
| 1.83 | H |
| 1.73 | G |
| 1.63 | L |

1. Null Hypothesis: all means are equal
2. Alternative Hypothesis: not all means equal simultaneously
3. Critical Value = 5.14
4. Test Statistic F = mSSG/mSSR

➢ Decision Rule

**Recall from lectures…**

<u>Some Background</u>:

- ANOVA looks at some measure of variation of the data taking all the data points together, and splits that up into two types.

a) Variation *between* the group means (a.k.a. actual differences between the groups, a signal)

b) Other variation, *within* the groups, put down to "errors" at this stage (a.k.a. noise)

<u>Remember</u>:

- The **more different the means**, the bigger the signal and the easier it is to see that they are actually different. This is **good for us**.

- The **more spread out the groups**, the bigger the noise and the harder it is to be certain where the means should be (*why? more spread* $\Rightarrow$ *larger* $s$ $\Rightarrow$ *wider confidence interval*). This is **not helpful** for us.

**Recall from lectures…**

<u>An Example</u>:

| Data | Group |
|---|---|
| 2 | A |
| 4 | A |
| 4 | B |
| 5 | B |
| 6 | B |
| 2 | C |
| 3 | C |
| 9 | C |
| 10 | C |

→

| Group | Data | | | |
|---|---|---|---|---|
| A | 2 | 4 | | |
| B | 4 | 5 | 6 | |
| C | 2 | 3 | 9 | 10 |

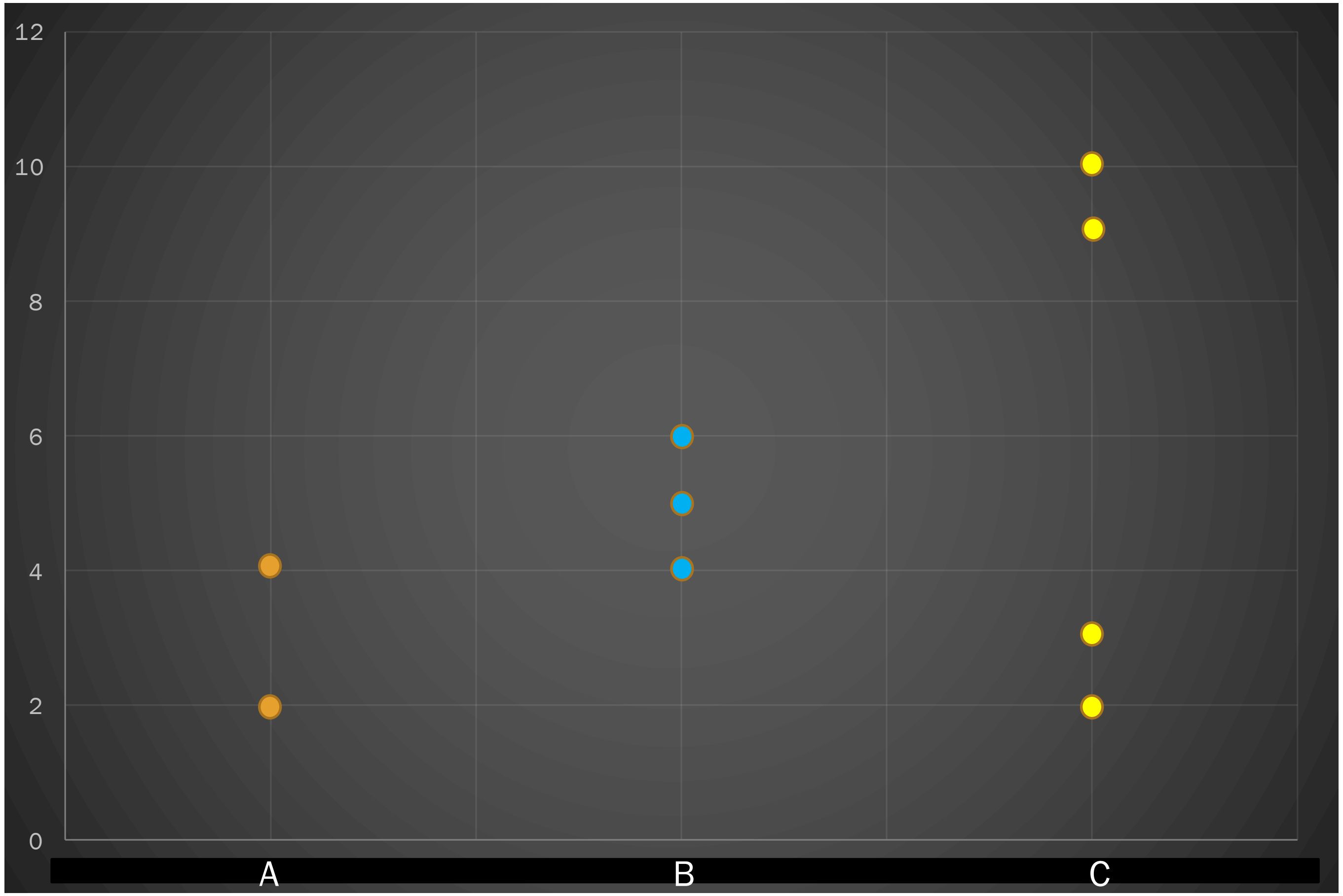
1. Our **null hypothesis** (our assumption) is that each group **should have the same mean**. That is:

$$H_0: \mu_A = \mu_B = \mu_C$$

2. Our **alternative** is that $H_0$ is not the case. Hence, $H_1$: $H_0$ *is not true*

Note, we might have $\mu_A = \mu_B \neq \mu_C$, or $\mu_A \neq \mu_B = \mu_C$, or $\mu_A = \mu_C \neq \mu_B$. All we would know is that they are not all the same.

## Recall from lectures…



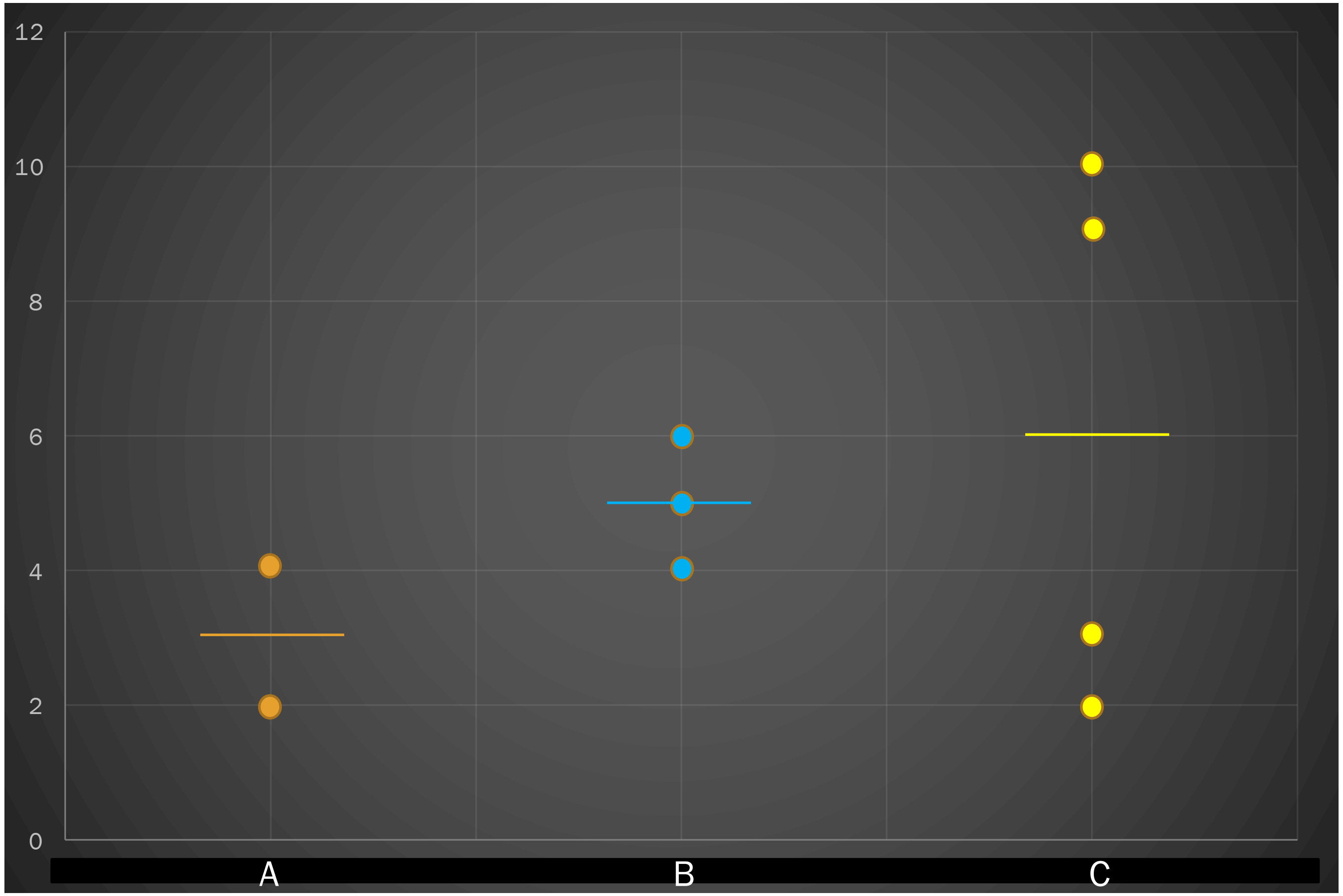

| Data | Group |
|---|---|
| 2 | A |
| 4 | A |
| 4 | B |
| 5 | B |
| 6 | B |
| 2 | C |
| 3 | C |
| 9 | C |
| 10 | C |

| Group | Data | | | |
|---|---|---|---|---|
| A | 2 | 4 | | |
| B | 4 | 5 | 6 | |
| C | 2 | 3 | 9 | 10 |

Signal

1. Group Mean
2. Grand Mean = 5
3. Group mean – grand mean

**Recall from lectures…**



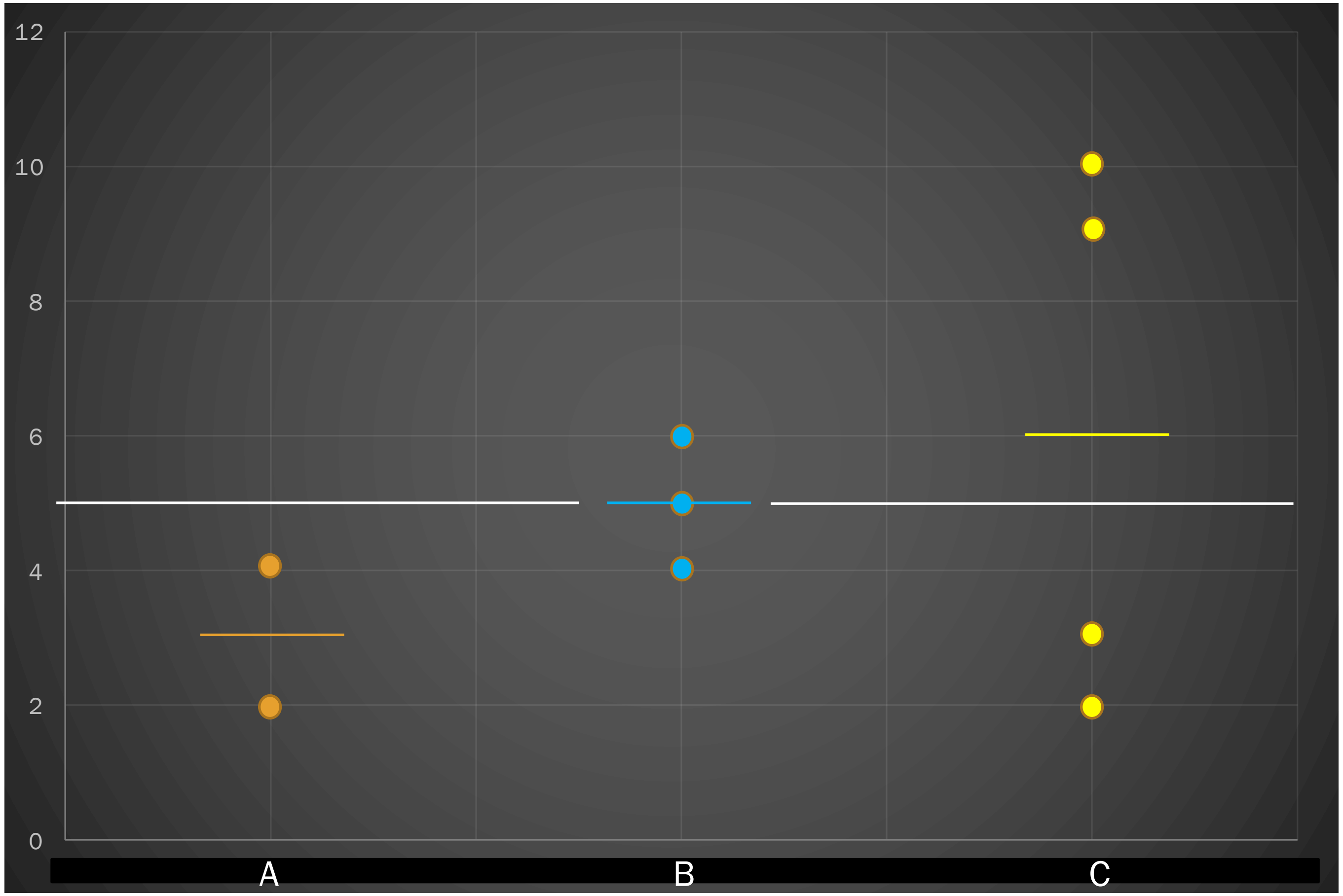

| Data | Group |
|---|---|
| 2 | A |
| 4 | A |
| 4 | B |
| 5 | B |
| 6 | B |
| 2 | C |
| 3 | C |
| 9 | C |
| 10 | C |

| Group | Data | | | |
|---|---|---|---|---|
| A | 2 | 4 | | |
| B | 4 | 5 | 6 | |
| C | 2 | 3 | 9 | 10 |

Signal

1. **Group Mean**
2. Grand Mean = 5
3. Group mean – grand mean

**Recall from lectures…**




| Data | Group |
|---|---|
| 2 | A |
| 4 | A |
| 4 | B |
| 5 | B |
| 6 | B |
| 2 | C |
| 3 | C |
| 9 | C |
| 10 | C |

| Group | Data | | | |
|---|---|---|---|---|
| A | 2 | 4 | | |
| B | 4 | 5 | 6 | |
| C | 2 | 3 | 9 | 10 |

Signal

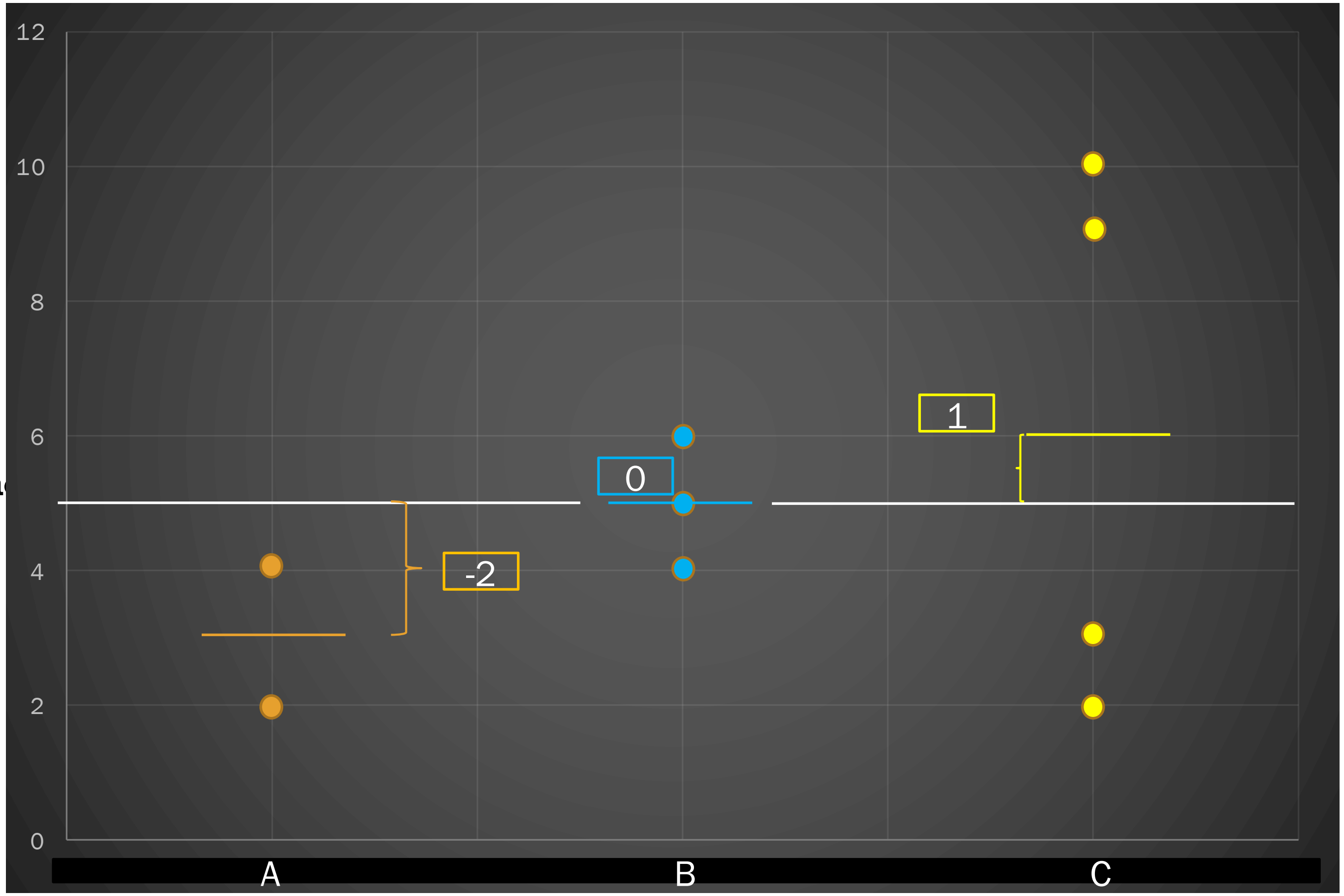
1. Group Mean
2. **Grand Mean = 5**
3. Group mean – grand mean

## Recall from lectures…

| Data | Group |
|---|---|
| 2 | A |
| 4 | A |
| 4 | B |
| 5 | B |
| 6 | B |
| 2 | C |
| 3 | C |
| 9 | C |
| 10 | C |

| Group | Data | | | |
|---|---|---|---|---|
| A | 2 | 4 | | |
| B | 4 | 5 | 6 | |
| C | 2 | 3 | 9 | 10 |

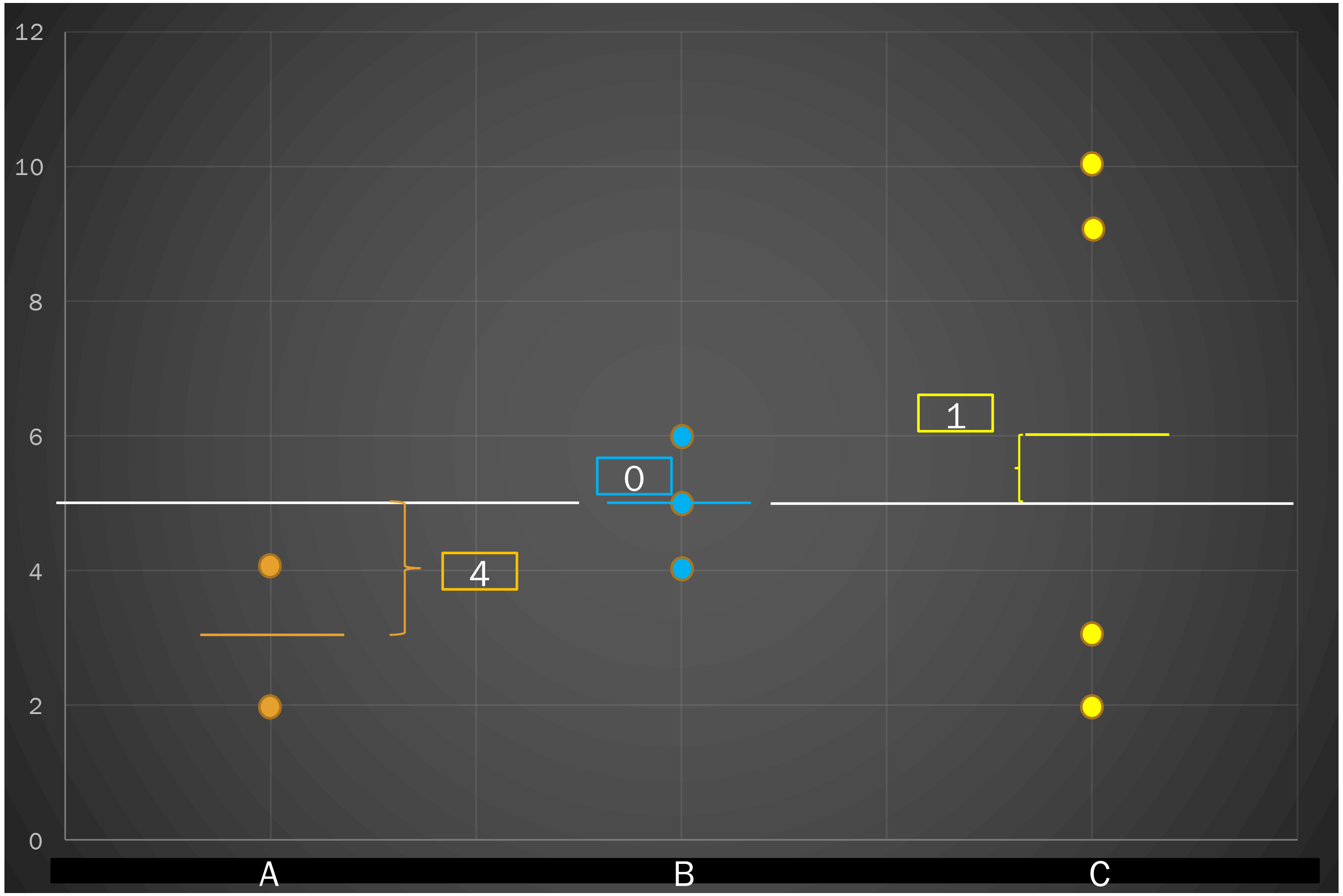
m




<u>Signal</u>

1. Group Mean
2. Grand Mean = 5
3. **Group mean – grand**

## Recall from lectures...



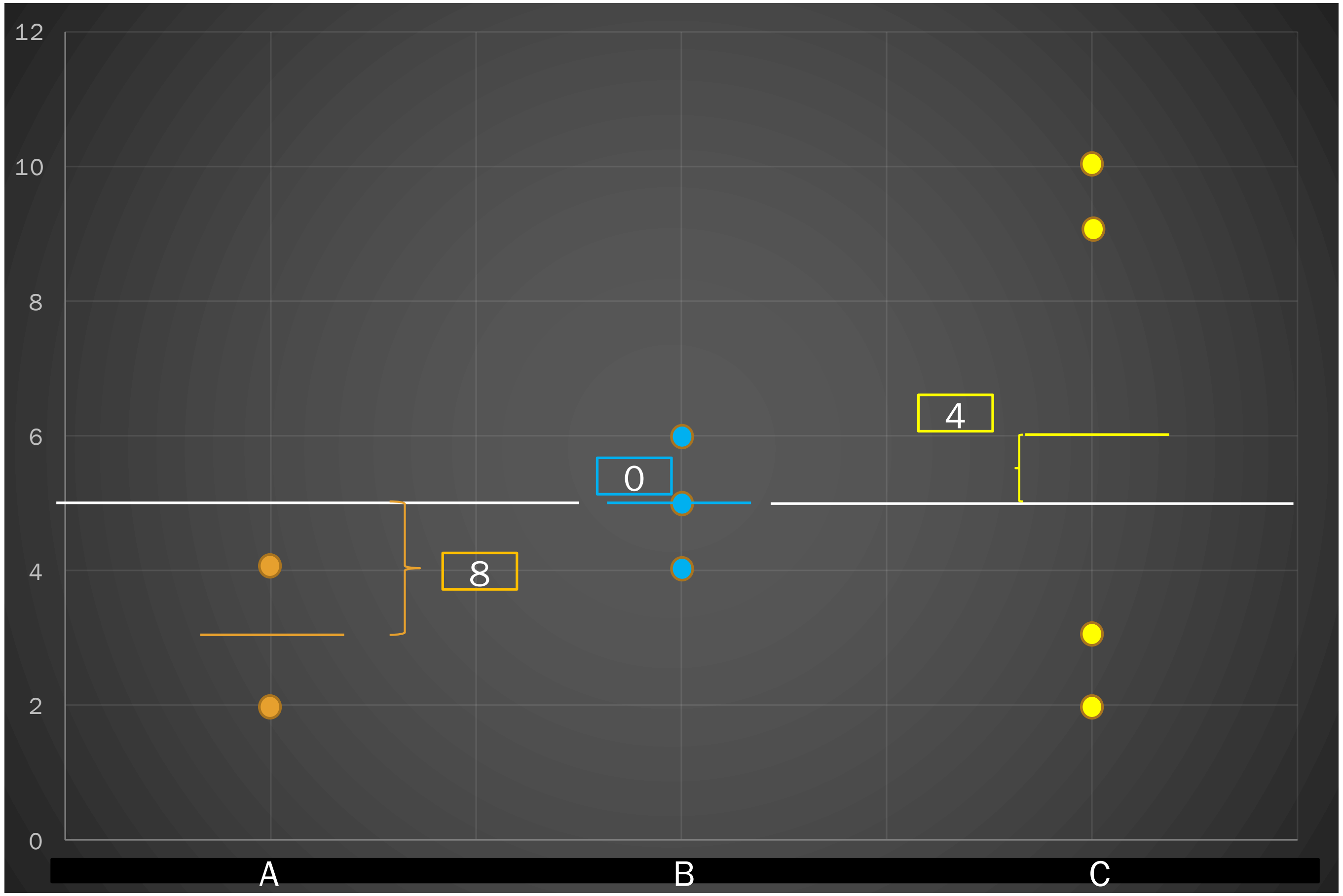

| Data | Group |
|---|---|
| 2 | A |
| 4 | A |
| 4 | B |
| 5 | B |
| 6 | B |
| 2 | C |
| 3 | C |
| 9 | C |
| 10 | C |

| Group | Data | | | |
|---|---|---|---|---|
| A | 2 | 4 | | |
| B | 4 | 5 | 6 | |
| C | 2 | 3 | 9 | 10 |

Signal

1. Group Mean
2. Grand Mean = 5
3. Group mean – grand mean
4. **Square the distances**

## Recall from lectures…



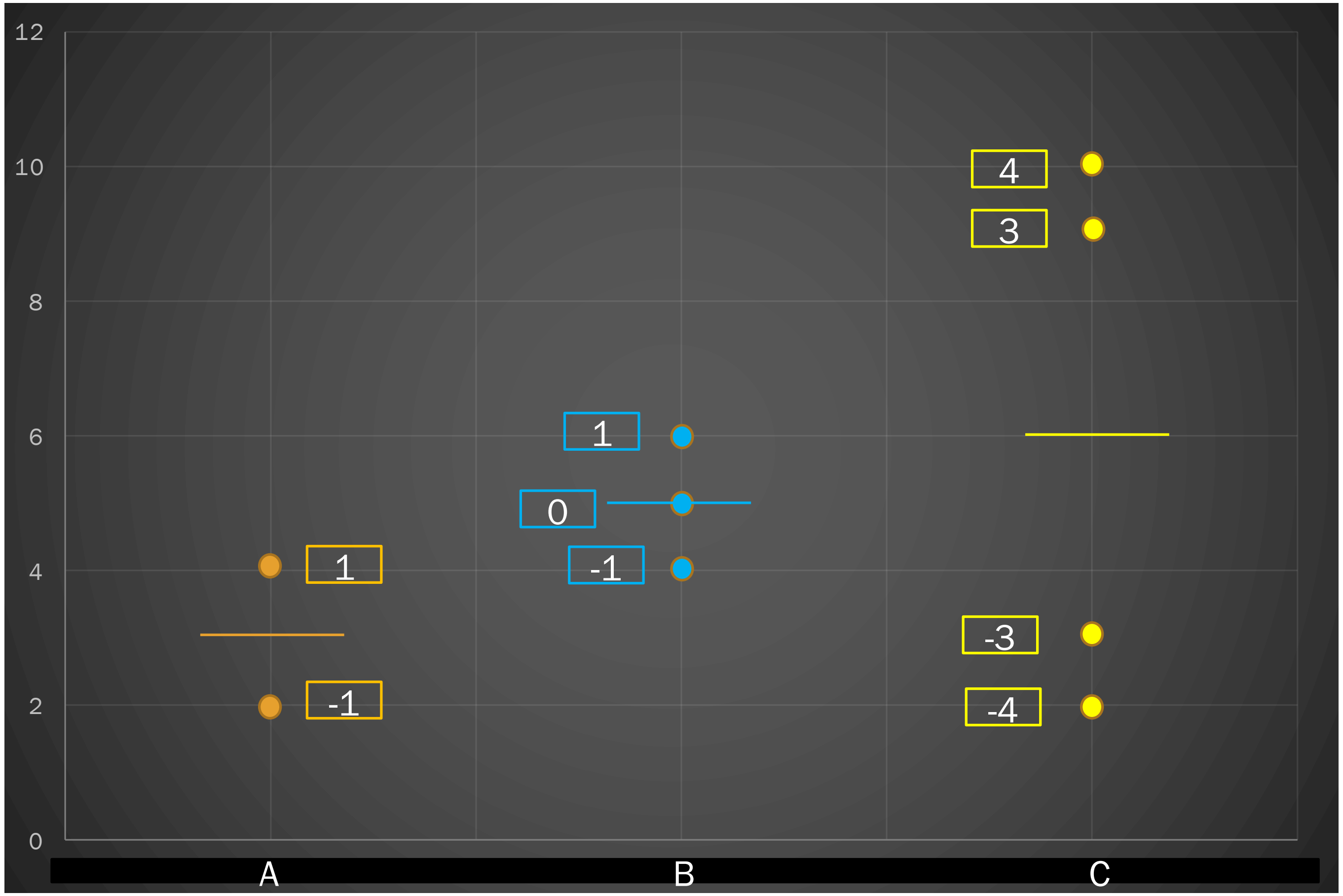

| Data | Group |
|---|---|
| 2 | A |
| 4 | A |
| 4 | B |
| 5 | B |
| 6 | B |
| 2 | C |
| 3 | C |
| 9 | C |
| 10 | C |

| Group | Data | | | |
|---|---|---|---|---|
| A | 2 | 4 | | |
| B | 4 | 5 | 6 | |
| C | 2 | 3 | 9 | 10 |

Signal

1. Group Mean
2. Grand Mean = 5
3. Group mean – grand mean
4. Square the distances
5. take each **squared distance** for each datapoint, that’s twice, three times, and four
6. Sum to get signal = 12

## Recall from lectures...



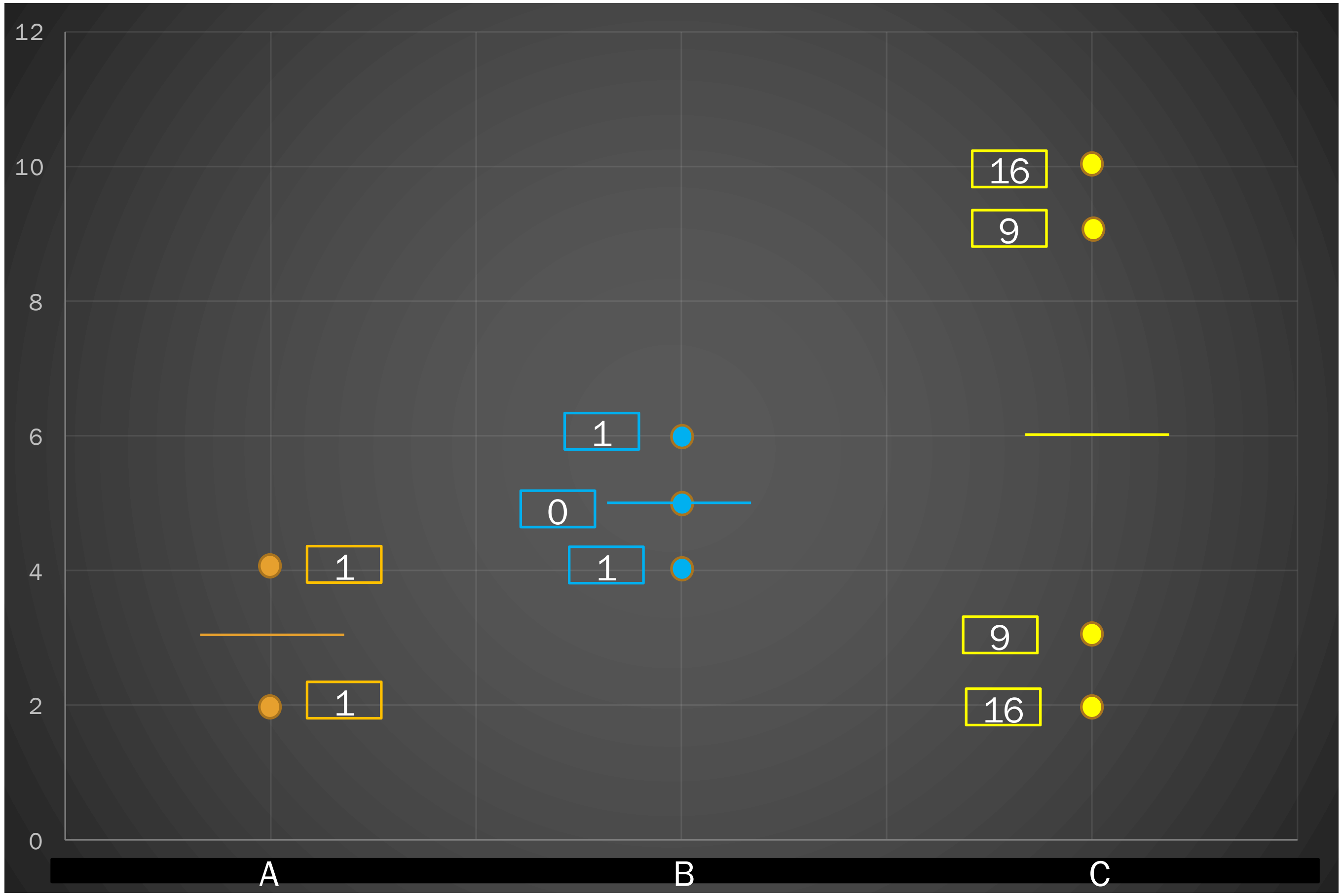

| Data | Group |
|---|---|
| 2 | A |
| 4 | A |
| 4 | B |
| 5 | B |
| 6 | B |
| 2 | C |
| 3 | C |
| 9 | C |
| 10 | C |

| Group | Data | | | |
|---|---|---|---|---|
| A | 2 | 4 | | |
| B | 4 | 5 | 6 | |
| C | 2 | 3 | 9 | 10 |

Noise

1. Look at the **distances** from the group mean
2. Square the distances

**Recall from lectures...**



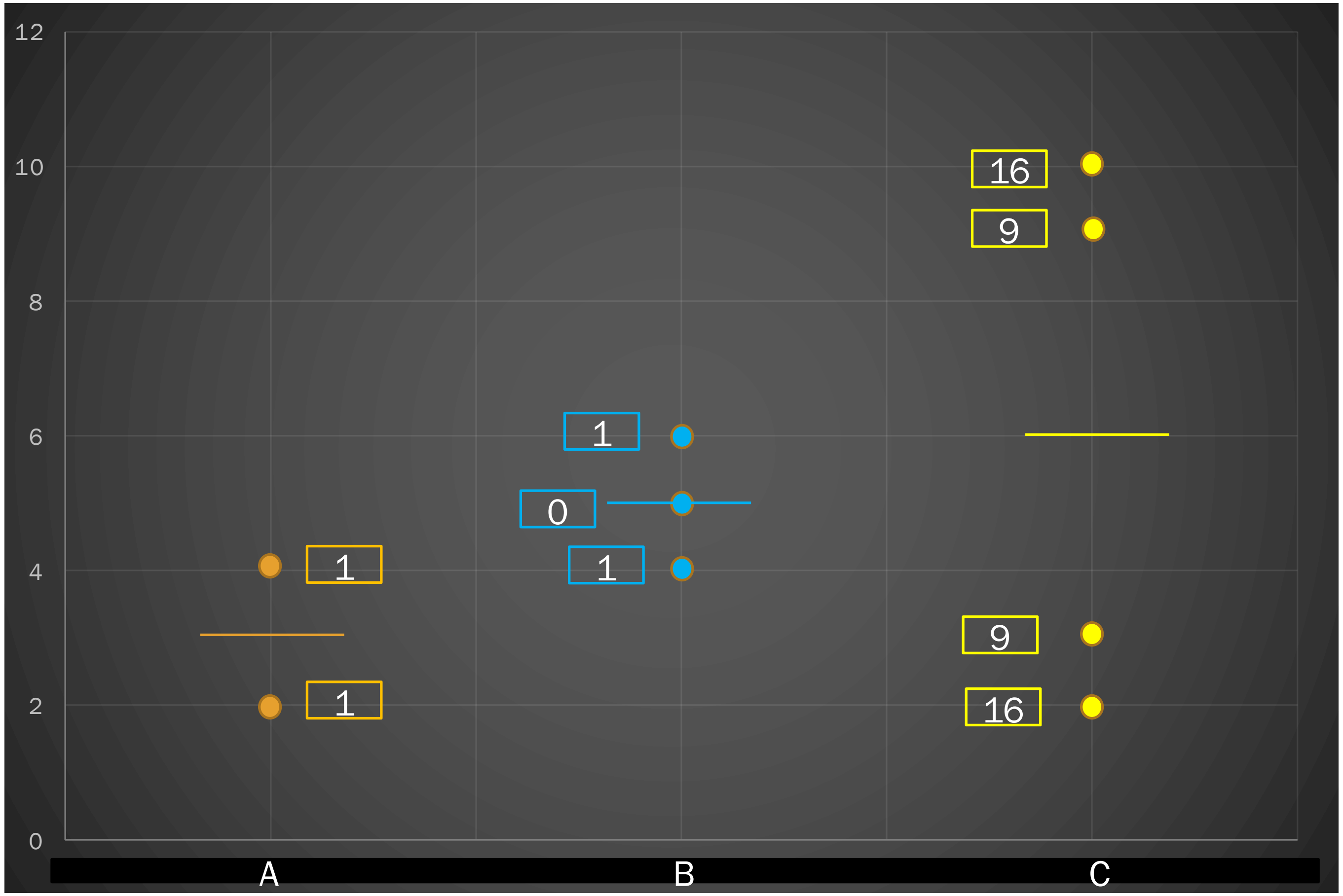

| Data | Group |
|---|---|
| 2 | A |
| 4 | A |
| 4 | B |
| 5 | B |
| 6 | B |
| 2 | C |
| 3 | C |
| 9 | C |
| 10 | C |

| Group | Data | | | |
|---|---|---|---|---|
| A | 2 | 4 | | |
| B | 4 | 5 | 6 | |
| C | 2 | 3 | 9 | 10 |

Noise

1. Look at the distances from the group mean
2. **Square the distances**

**Recall from lectures…**




| Data | Group |
|---|---|
| 2 | A |
| 4 | A |
| 4 | B |
| 5 | B |
| 6 | B |
| 2 | C |
| 3 | C |
| 9 | C |
| 10 | C |

| Group | Data | | | |
|---|---|---|---|---|
| A | 2 | 4 | | |
| B | 4 | 5 | 6 | |
| C | 2 | 3 | 9 | 10 |

<u>Noise</u>

1. Look at the distances from the group mean
2. Square the distances
3. **Sum to get noise = 54**

**Recall from lectures…**

<u>An Example</u>:

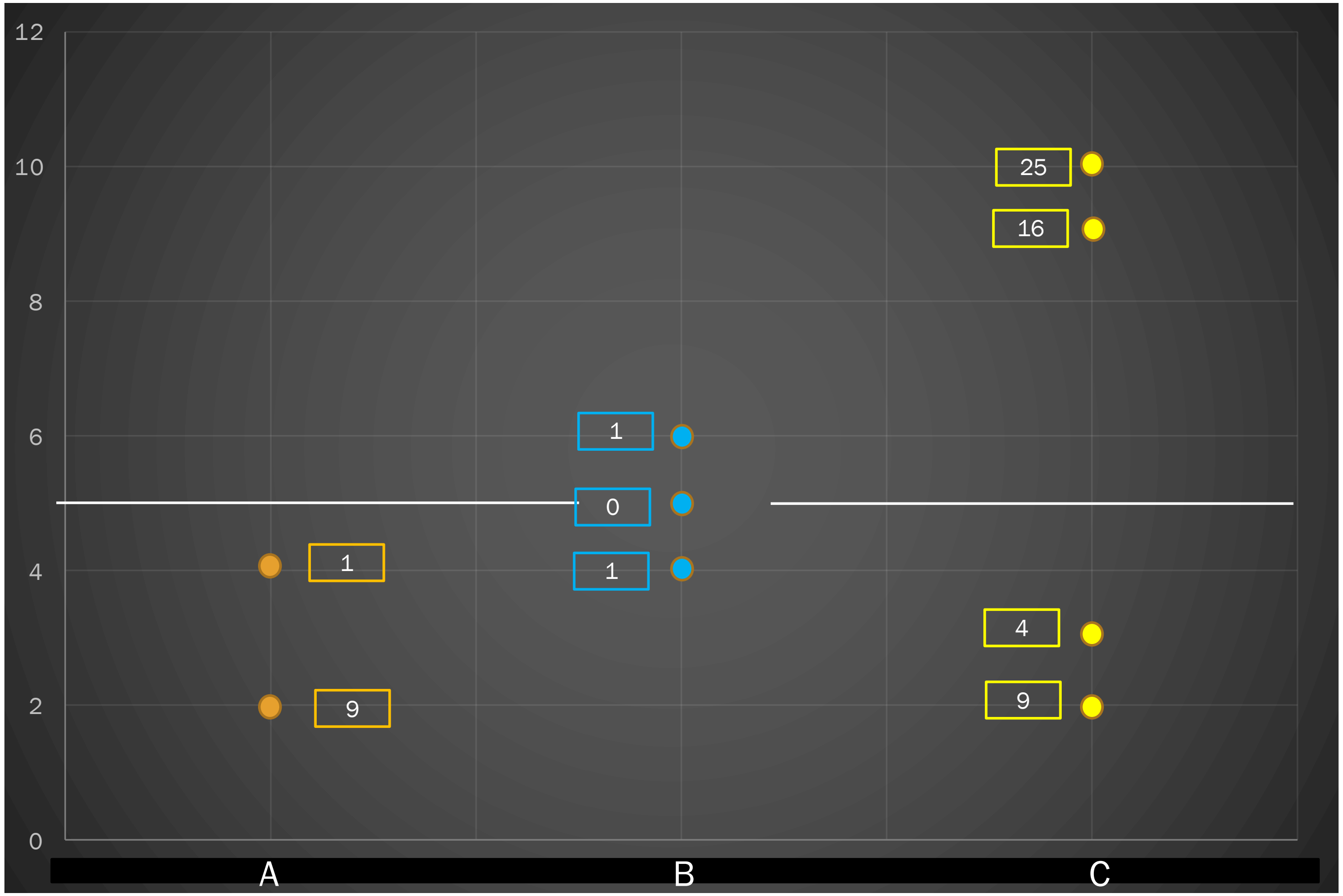
| Data | Group |
|---|---|
| 2 | A |
| 4 | A |
| 4 | B |
| 5 | B |
| 6 | B |
| 2 | C |
| 3 | C |
| 9 | C |
| 10 | C |

| Group | Data | | | |
|---|---|---|---|---|
| A | 2 | 4 | | |
| B | 4 | 5 | 6 | |
| C | 2 | 3 | 9 | 10 |

Hence, **Total** = Signal + Noise = 12 + 54 = 66.

❖ *We can also calculate directly the total, that is the sum of the squared distances from the grand mean*

**Recall from lectures…**




| Data | Group |
|---|---|
| 2 | A |
| 4 | A |
| 4 | B |
| 5 | B |
| 6 | B |
| 2 | C |
| 3 | C |
| 9 | C |
| 10 | C |

| Group | Data | | | |
|---|---|---|---|---|
| A | 2 | 4 | | |
| B | 4 | 5 | 6 | |
| C | 2 | 3 | 9 | 10 |

## Total

1. distance from grand mean
2. square the distances, sum
3. Total = 66

**Recall from lectures…**

| Source | SS | df | MSS | F |
|---|---|---|---|---|
| Groups (Signal) | 12 | | | |
| Errors (Noise) | 54 | | | |
| Total | 66 | | | |

Which doesn't look good for the signal.

- I have three groups, thus I have 2 degrees of freedom
- I have 2 datapoints in group A, so can set 1 where I want: 1 df. I also have 3 in group B, so can set 2, so 2 df. Finally, I also have 4 in C, so can set 3, so 3 df.

**Recall from lectures…**

| Source | SS | df | MSS | F |
|---|---|---|---|---|
| Groups (Signal) | 12 | 2 | | |
| Errors (Noise) | 54 | 6 | | |
| Total | 66 | | | |

Which doesn't look good for the signal.

- I have three groups, thus I have 2 degrees of freedom
- I have 2 datapoints in group A, so can set 1 where I want: 1 df. I also have 3 in group B, so can set 2, so 2 df. Finally, I also have 4 in C, so can set 3, so 3 df.
- Total = 8 (we already know that; since we have 9 datapoints in total, n-1 = 8)

**Recall from lectures…**

| Source | SS | df | MSS | F |
|---|---|---|---|---|
| Groups (Signal) | 12 | 2 | | |
| Errors (Noise) | 54 | 6 | | |
| Total | 66 | 8 | | |

Which doesn't look good for the signal.

- I have three groups, thus I have 2 degrees of freedom
- I have 2 datapoints in group A, so can set 1 where I want: 1 df. I also have 3 in group B, so can set 2, so 2 df. Finally, I also have 4 in C, so can set 3, so 3 df.
- Total = 8 (we already know that; since we have 9 datapoints in total, n-1 = 8)
- We divide our measures (the Sums of Squares (SS)) by the degrees of freedom. This gives the Mean Sums of Squares (MSS), a better measure of signal and noise.

**Recall from lectures…**

| Source | SS | df | MSS | F |
|---|---|---|---|---|
| Groups (Signal) | 12 | 2 | 6 | |
| Errors (Noise) | 54 | 6 | 9 | |
| Total | 66 | 8 | | |

Which doesn't look good for the signal.

- I have three groups, thus I have 2 degrees of freedom
- I have 2 datapoints in group A, so can set 1 where I want: 1 df. I also have 3 in group B, so can set 2, so 2 df. Finally, I also have 4 in C, so can set 3, so 3 df.
- Total = 8 (we already know that; since we have 9 datapoints in total, n-1 = 8)
- We divide our measures (the Sums of Squares (SS)) by the degrees of freedom. This gives the Mean Sums of Squares (MSS), a better measure of signal and noise.
- We compare the signal to the noise. How? MSSsignal/MSSnoise

**Recall from lectures…**

| Source | SS | df | MSS | F |
|---|---|---|---|---|
| Groups (Signal) | 12 | 2 | 6 | 0.667 |
| Errors (Noise) | 54 | 6 | 9 | |
| Total | 66 | 8 | | |

Which doesn't look good for the signal.

- I have three groups, thus I have 2 degrees of freedom
- I have 2 datapoints in group A, so can set 1 where I want: 1 df. I also have 3 in group B, so can set 2, so 2 df. Finally, I also have 4 in C, so can set 3, so 3 df.
- Total = 8 (we already know that; since we have 9 datapoints in total, n-1 = 8)
- We divide our measures (the Sums of Squares (SS)) by the degrees of freedom. This gives the Mean Sums of Squares (MSS), a better measure of signal and noise.
- We compare the signal to the noise. How? MSSsignal/MSSnoise
- This process gives us the test statistic, called F!
- We compare the TS with critical value to decide if we reject the null or not.

## Recall from lectures...

In practise: SPSS output!

**ANOVA**

Height (m)

| | Sum of Squares | df | Mean Square | F | Sig. |
|---|---|---|---|---|---|
| Between Groups | .117 | 7 | .017 | 1.040 | .406 |
| Within Groups | 2.427 | 151 | .016 | | |
| Total | 2.544 | 158 | | | |

**Question 1**

ANOVA looks at:

1. the difference of means over a standard deviation.

2. the (weighted) variance between groups compared to the (weighted) variance within groups.

3. whether a coefficient is significantly different from 0.

4. two continuous variables against a straight line.

**Question 1**

ANOVA looks at:

1. the difference of means over a standard deviation.

2. **the (weighted) variance between groups compared to the (weighted) variance within groups**.

3. whether a coefficient is significantly different from 0.

4. two continuous variables against a straight line.

## Question 2

A piece of research produces the following data:

| | | | | | |
|---|---|---|---|---|---|
| Group A | 31 | 15 | 14 | 21 | 35 |
| Group B | 26 | 25 | 43 | | |
| Group C | 29 | 37 | 19 | 16 | |

What is the contribution to SSE (the Sum of Squares Errors - also known as "Within Groups") of Group C?

**Question 2**

A piece of research produces the following data:

| | | | | | |
|---|---|---|---|---|---|
| Group A | 31 | 15 | 14 | 21 | 35 |
| Group B | 26 | 25 | 43 | | |
| Group C | 29 | 37 | 19 | 16 | |

What is the contribution to SSE (the Sum of Squares Errors - also known as "Within Groups") of Group C?

We focus on group C. We need $n\sigma^2$

**Question 2**

A piece of research produces the following data:

| | | | | | |
|---|---|---|---|---|---|
| Group A | 31 | 15 | 14 | 21 | 35 |
| Group B | 26 | 25 | 43 | | |
| Group C | 29 | 37 | 19 | 16 | |

What is the contribution to SSE (the Sum of Squares Errors - also known as "Within Groups") of Group C?

We focus on group C. We need $n\sigma^2$ for C

1. Calculate the variance: $\sigma^2 = 69.1875$
2. $n\sigma^2 = 4 * 69.187 = 276.75$

- Note that for the overall SSE we do the same for A and B and then sum

## Question 3

A piece of research produces the following data:

| | | | | |
|---|---|---|---|---|
| Group A | 2 | 3 | 9 | 10 |
| Group B | 2 | 4 | | |
| Group C | 4 | 5 | 6 | |

What is SSG (the Sum of Squares Groups - also known as "Between Groups")?

## Question 3

A piece of research produces the following data:

| | | | | |
|---|---|---|---|---|
| Group A | 2 | 3 | 9 | 10 |
| Group B | 2 | 4 | | |
| Group C | 4 | 5 | 6 | |

What is SSG (the **Sum of Squares Groups** - also known as "**Between Groups**")?

1. Calculate **the mean of each group**:

## Question 3

A piece of research produces the following data:

| | | | | |
|---|---|---|---|---|
| Group A | 2 | 3 | 9 | 10 |
| Group B | 2 | 4 | | |
| Group C | 4 | 5 | 6 | |

What is SSG (the **Sum of Squares Groups** - also known as "**Between Groups**")?

1. Calculate the mean of each group: $\bar{x}_A = 6$, $\bar{x}_B = 3$, $\bar{x}_C = 5$
2. Calculate the variance from:

| $\bar{x}$ | frequency |
|---|---|
| 6 | 4 |
| 3 | 2 |
| 5 | 3 |

$\sigma^2 =$

## Question 3

A piece of research produces the following data:

| | | | | |
|---|---|---|---|---|
| Group A | 2 | 3 | 9 | 10 |
| Group B | 2 | 4 | | |
| Group C | 4 | 5 | 6 | |

What is SSG (the Sum of Squares Groups - also known as "Between Groups")?

1. Calculate the mean of each group: $\bar{x}_A = 6$, $\bar{x}_B = 3$, $\bar{x}_C = 5$
2. Calculate the variance from:

| $\bar{x}$ | frequency |
|---|---|
| 6 | 4 |
| 3 | 2 |
| 5 | 3 |

$\sigma^2 = 1.33$

3. Calculate $n\sigma^2$ =

## Question 3

A piece of research produces the following data:

| | | | | |
|---|---|---|---|---|
| Group A | 2 | 3 | 9 | 10 |
| Group B | 2 | 4 | | |
| Group C | 4 | 5 | 6 | |

What is SSG (the Sum of Squares Groups - also known as "Between Groups")?

1. Calculate the mean of each group: $\bar{x}_A = 6$, $\bar{x}_B = 3$, $\bar{x}_C = 5$
2. Calculate the variance from:

| $\bar{x}$ | frequency |
|---|---|
| 6 | 4 |
| 3 | 2 |
| 5 | 3 |

$\sigma^2 = 1.33$

3. Calculate $n\sigma^2 = 9*1.33 = 11.97$

## Question 4

A piece of research produces the following data:

| | | | | | |
|---|---|---|---|---|---|
| Group A | 21 | 19 | 39 | 17 | 48 |
| Group B | 28 | 44 | 20 | | |
| Group C | 35 | 39 | 23 | 32 | |

What is SST (the Sum of Squares Total)?

## Question 4

A piece of research produces the following data:

| | | | | | |
|---|---|---|---|---|---|
| Group A | 21 | 19 | 39 | 17 | 48 |
| Group B | 28 | 44 | 20 | | |
| Group C | 35 | 39 | 23 | 32 | |

What is SST (the Sum of Squares **Total**)?

- $n\sigma^2 =$

## Question 4

A piece of research produces the following data:

| | | | | | |
|---|---|---|---|---|---|
| Group A | 21 | 19 | 39 | 17 | 48 |
| Group B | 28 | 44 | 20 | | |
| Group C | 35 | 39 | 23 | 32 | |

What is SST (the Sum of Squares Total)?

- $n\sigma^2$ = 12*102.743 = 1,232.917

## Question 5

A piece of research produces the following data:

| | | | | | |
|---|---|---|---|---|---|
| Group A | 36 | 17 | 27 | 13 | 20 |
| Group B | 19 | 28 | 43 | | |
| Group C | 22 | 33 | 34 | 24 | |

What is the Test Statistic?

## Question 5

A piece of research produces the following data:

| | | | | | |
|---|---|---|---|---|---|
| Group A | 36 | 17 | 27 | 13 | 20 |
| Group B | 19 | 28 | 43 | | |
| Group C | 22 | 33 | 34 | 24 | |

What is the Test Statistic?

- Remember: $TS = \frac{SSG/\mathrm{d}f_G}{SSE/\mathrm{d}f_\mathrm{E}}$, where SSG is Sum of Squares Groups, SSE Sum of Squares Errors, $\mathrm{d}f_G$ is one fewer the number of groups, and $\mathrm{d}f_E$ is the number of datapoints minus the number of groups.

**Question 5**

A piece of research produces the following data:

| | | | | | |
|---|---|---|---|---|---|
| Group A | 36 | 17 | 27 | 13 | 20 |
| Group B | 19 | 28 | 43 | | |
| Group C | 22 | 33 | 34 | 24 | |

What is the Test Statistic?

- Remember: $TS = \frac{SSG/\mathrm{d}f_G}{SSE/\mathrm{d}f_\mathrm{E}}$, where SSG is Sum of Squares Groups, SSE Sum of Squares Errors, $\mathrm{d}f_G$ is one fewer the number of groups, and $\mathrm{d}f_E$ is the number of datapoints minus the number of groups.
- To find SSE we need the $n\sigma^2$ of each group. Hence,
- $\sigma_\mathrm{A}^2 = ?$, $\sigma_B^2 = ?$, $\sigma_\mathrm{C}^2 = ?$

## Question 5

A piece of research produces the following data:

| | | | | | |
|---|---|---|---|---|---|
| Group A | 36 | 17 | 27 | 13 | 20 |
| Group B | 19 | 28 | 43 | | |
| Group C | 22 | 33 | 34 | 24 | |

What is the Test Statistic?

- Remember: $TS = \frac{SSG/df_G}{SSE/df_E}$, where SSG is Sum of Squares Groups, SSE Sum of Squares Errors, $df_G$ is one fewer the number of groups, and $df_E$ is the number of datapoints minus the number of groups.
- To find SSE we need the $n\sigma^2$ of each group. Hence,
- $\sigma_A^2 = 65.84$, $\sigma_B^2 = 98$, $\sigma_C^2 = 28.18$
- $n\sigma_A^2$ = ?, $n\sigma_B^2$ = ?, $n\sigma_C^2$ = ?. Hence. SSE = ?

## Question 5

A piece of research produces the following data:

| | | | | | |
|---|---|---|---|---|---|
| Group A | 36 | 17 | 27 | 13 | 20 |
| Group B | 19 | 28 | 43 | | |
| Group C | 22 | 33 | 34 | 24 | |

What is the Test Statistic?

- Remember: $TS = \frac{SSG/\mathrm{d}f_G}{SSE/\mathrm{d}f_\mathrm{E}}$, where SSG is Sum of Squares Groups, SSE Sum of Squares Errors, $\mathrm{d}f_G$ is one fewer the number of groups, and $\mathrm{d}f_E$ is the number of datapoints minus the number of groups.
- To find SSE we need the $n\sigma^2$ of each group. Hence,
- $\sigma_\mathrm{A}^2 = 65.84$, $\sigma_B^2 = 98$, $\sigma_\mathrm{C}^2 = 28.18$
- $n\sigma_\mathrm{A}^2 = 329.2$, $n\sigma_B^2 = 294$, $n\sigma_C^2 = 112.72$. Hence. SSE = 735.92
- $\mathrm{d}f_E = 12 - 3 = 9$, $\mathrm{d}f_G = 3 - 1 = 2$

## Question 5

A piece of research produces the following data:

| | | | | | |
|---|---|---|---|---|---|
| Group A | 36 | 17 | 27 | 13 | 20 |
| Group B | 19 | 28 | 43 | | |
| Group C | 22 | 33 | 34 | 24 | |

What is the Test Statistic?

- Remember: $TS = \frac{SSG/\mathrm{d}f_G}{SSE/\mathrm{d}f_E}$, where SSG is Sum of Squares Groups, SSE Sum of Squares Errors, $\mathrm{d}f_G$ is one fewer the number of groups, and $\mathrm{d}f_E$ is the number of datapoints minus the number of groups.
- SSE = 735.92, $\mathrm{d}f_E$ = 12 – 3 = 9, $\mathrm{d}f_G$ = 2
- For the SSG, we need to calculate the mean of each group: $\bar{x}_A = 22.6$, $\bar{x}_B = 30$, $\bar{x}_C = 28.25$. Then, we calculate $n\sigma^2$ of these three datapoints with frequencies 5, 3, and 4 respectively.

## Question 5

A piece of research produces the following data:

| | | | | | |
|---|---|---|---|---|---|
| Group A | 36 | 17 | 27 | 13 | 20 |
| Group B | 19 | 28 | 43 | | |
| Group C | 22 | 33 | 34 | 24 | |

What is the Test Statistic?

- Remember: $TS = \frac{SSG/\mathrm{d}f_G}{SSE/\mathrm{d}f_\mathrm{E}}$, where SSG is Sum of Squares Groups, SSE Sum of Squares Errors, $\mathrm{d}f_G$ is one fewer the number of groups, and $\mathrm{d}f_E$ is the number of datapoints minus the number of groups.
- SSE = 735.92, $\mathrm{d}f_E$ = 12 – 3 = 9, $\mathrm{d}f_G$ = 2
- For the SSG, we need to calculate the mean of each group: $\bar{x}_A = 22.6$, $\bar{x}_B = 30$, $\bar{x}_C = 28.25$. Then, we calculate $n\sigma^2$ of these three datapoints with frequencies 5, 3, and 4 respectively.
- SSG = ?

| X | frequency |
|---|---|
| 22.6 | 5 |
| 30 | 3 |
| 28.25 | 4 |

## Question 5

A piece of research produces the following data:

| | | | | | |
|---|---|---|---|---|---|
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| X | frequency |
|---|---|
| 22.6 | 5 |
| 30 | 3 |
| 28.25 | 4 |

- SSG = 124.716
- TS =

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| X | frequency |
|---|---|
| 22.6 | 5 |
| 30 | 3 |
| 28.25 | 4 |

- SSG = 124.716
- $\mathrm{TS} = \frac{124.716/2}{735.92/9} = 0.7626$

## Question 6

Which of the following is a correct interpretation of the following SPSS output?

**ANOVA**

VAR00001

| | Sum of Squares | df | Mean Square | F | Sig. |
|---|---|---|---|---|---|
| Between Groups | 40.100 | 2 | 20.050 | .279 | .759 |
| Within Groups | 1580.540 | 22 | 71.843 | | |
| Total | 1620.640 | 24 | | | |

1. There were 24 datapoints.
2. Because 0.759 > 0.279, we reject the null hypothesis and conclude that at least one group is significantly different from the rest.
3. There were 22 groups
4. We fail to reject the null hypothesis - no group seems to be significantly different

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**ANOVA**

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| Within Groups | 1580.540 | 22 | 71.843 | | |
| Total | 1620.640 | 24 | | | |

1. There were 24 datapoints. (25 datapoints)
2. Because 0.759 > 0.279, we reject the null hypothesis and conclude that at least one group is significantly different from the rest.
3. There were 22 groups (3 groups)
4. **We fail to reject the null hypothesis - no group seems to be significantly different**

## Sum up

1. SSE: we focus on each group $\Rightarrow$ find the variance of each group, then multiply by the sample size of the group $\Rightarrow$ sum all contributions

2. SSG: between groups spread $\Rightarrow$ find the mean of each group $\Rightarrow$ construct a table of the group means and the frequency of each mean $\Rightarrow$ calculate the variance of this table $\Rightarrow$ multiply the variance with the total number of datapoints

3. SST: total variation $\Rightarrow$ find the variance of all datapoints and multiply by the number of datapoints

4. Test Statistic

[redacted]